

Deepak Singh

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Summary

Lead Data Engineer with 9.8 years of experience specializing in **enterprise data strategy, lakehouse governance, and cloud-native distributed data systems** across **FinTech, Retail, Cybersecurity, HR, Telecom, and Healthcare domains, including Global Capability Center (GCC) environments**. Proven track record of designing **high-throughput batch and real-time data platforms** using **AWS, Azure, Databricks, and PySpark**. Experienced in translating complex business requirements into scalable, high-availability data infrastructure, driving **cost optimization**, and leading cross-functional engineering teams. Experienced in integrating **generative AI tools** such as **GitHub Copilot, Windsurf, and AWS Bedrock** to improve SDLC efficiency and platform reliability.

Experience

Lead Data Engineer

7-Eleven | Bangalore | May 2025 – June 2026

Project: Retail Data Model

- **Drove end-to-end platform ownership across enterprise-scale retail data platform architecture, delivery, and lifecycle governance** while leading a cross-functional team of 7 data engineers across solution design, code reviews, and release planning.
- **Defined enterprise data strategy and architected scalable dimensional data models and lakehouse architecture** for multi-billion-record sales datasets, enabling governed, high-performance analytics across multiple retail business domains.
- Designed ingestion and transformation frameworks for ServiceNow operational data, improving incident analytics and operational visibility.
- Strengthened data reliability by integrating **Monte Carlo-based observability and anomaly detection, reducing production data incidents by 40%** and improving SLA adherence.
- Spearheaded **AI-assisted engineering initiatives using Windsurf and GitHub Copilot** to accelerate software delivery, improve code quality, and enhance engineering productivity.
- Architected and developed a scalable **Databricks-based chargeback dispute analytics platform** that consolidated multi-source transaction data and automated evidence generation for **Stripe and American Express disputes**, improving **dispute win rates by 35%** and **protecting \$2M in revenue**.

Senior Data Engineer

KnowBe4 | Bangalore | August 2024 – April 2025

Project: Cybersecurity Risk Analytics Platform

- **Led the architecture and development of cloud-native ETL pipelines for user behavior analytics and risk scoring**, enabling proactive detection of phishing-risk patterns and driving targeted training that **reduced employee susceptibility by 86% and breach risk by 25%**.
- Designed and implemented **event-driven architecture** leveraging Lambda, Step Functions, and EMR Serverless for large-scale user behavior analytics, improving scalability and operational efficiency.
- Designed a **generative AI-based proof of concept using AWS Bedrock for automated unit test generation** and AI-assisted pull request creation, improving developer productivity.
- **Optimized CI/CD release pipelines** by integrating schema migrations and automated deployment validation, improving deployment reliability and reducing manual release effort.
- Mentored engineering teams on distributed data processing, cloud-native architecture, and modern data engineering practices.

Senior Data Engineer

Accenture | Bangalore | March 2020 – July 2024

Client: **GAP** | Retail Data Platform

- **Owned technical strategy and stakeholder alignment for enterprise retail clients**, translating business needs into scalable architecture on Databricks across engineering and platform teams.
- **Architected fault-tolerant PySpark pipelines on Databricks** to process terabytes of streaming sales data daily with high availability across production workloads.
- Led performance optimization initiatives that **reduced data processing time by 40%** and **client infrastructure costs by 30%** through advanced Spark tuning, resource allocation strategies, and pipeline re-architecture, establishing performance benchmarks adopted across the team.

- **Implemented data quality monitoring and alerting using Lightup**, significantly improving data integrity and reducing data anomalies.
- Established **lakehouse governance standards by integrating Unity Catalog** for metadata management and workspace-level access control, improving collaboration, security, and audit-readiness across teams.

Client: Verana Health | Clinical Prescription Recommendation Platform

- **Architected and scaled the core healthcare data platform** on AWS, engineering optimized distributed workloads and **Airflow orchestrations** to process large-scale clinical prescription data.
- Partnered with business and data science teams to design frameworks transforming complex prescription data for **predictive clinical recommendation engines**.
- Conducted proof of concepts for **MongoDB Global Clusters** and **MongoDB Kafka Connector** integration.

Member of Technical Staff II

Mavenir | Bangalore | January 2018 – March 2020

Project: Telecom Service Assurance Data Platform

- Architected data ingestion pipelines to extract high-volume telecom **IMS session data** and **publish it to Kafka**, enabling service assurance platforms to monitor call success and server latency.
- Developed microservices-based APIs using FastAPI to support downstream operational integrations.
- Increased code maintainability and deployment quality by **integrating SonarQube coverage** validation into CI/CD pipelines.

Associate Software Engineer

Huawei | Bangalore | September 2016 – December 2017

Project: Telecom Revenue Assurance & Analytics Platform

- Engineered scalable ETL pipelines processing **millions of Call Detail Records (CDRs)** daily to support telecom charging, billing, and inter-operator roaming settlement.
- Integrated processed telecom data with **Enterprise Data Warehouse (EDW)** and convergent billing systems, enabling downstream analytics, reporting, and financial reconciliation.

Technical Skills

Languages: Python, SQL, OOP, C++

Cloud Platforms: Azure, AWS

Databases: MySQL, MongoDB, DynamoDB

Web Services: REST, FastAPI, Django, Azure Web App

Big Data Tools: PySpark, Hadoop, Kafka, Databricks, SonarQube, Splunk, Datadog, Azure Data Lake Storage (ADLS), Unity Catalog, Google Cloud Storage (GCS), Google Composer, Key Vault, Hive, Lightup, Monte Carlo, EMR, Amazon EventBridge, Lambda, Glue, Athena, Step Functions, S3, SQS

Data Engineering Concepts: Data Quality, Data Governance, Medallion Architecture, CDC, SCD, Data Mesh, SDK Framework, metadata-driven pipelines

Orchestration: Databricks Workflows, Airflow, ADF

File Formats: Avro, CSV, JSON, Parquet, ASN.1 (binary file format), and Delta

CI/CD: GitLab, Terraform

AI Tools: GitHub Copilot, Windsurf, AWS Bedrock, Generative AI

Education

Uttarakhand Technical University | B.Tech in ECE | August 2012 – June 2016

Certifications

Databricks Associate Developer for Spark 3.0 | Azure AZ-900 | GCP Associate Cloud Engineer

Awards & Recognition

Special Contribution Award (7-Eleven) | Recognized for contributions to the data module (Accenture) | Recognized as Best Developer (Huawei)